FATIMA JINNAH WOMEN UNIVERSITY

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Course: Web Engineering

Section: BCS-5(B)

Project Report

Project Title:

BOOKHIVE: Library Management System for Librarian

1. Introduction

BookHive is a web-based Library Management System developed to automate and streamline traditional library operations. The system is primarily designed for librarians, enabling them to manage books, members, issuing/returning of books, and reports through a centralized digital platform.

This project is developed as part of the Web Engineering course, focusing on requirement analysis, system design, implementation, and documentation. The application follows standard web engineering practices including modular development, database normalization, and separation of concerns.

The system is implemented using HTML, CSS, JavaScript, Bootstrap, PHP, and MySQL, developed in Visual Studio Code, and deployed locally using XAMPP.

2. Problem Statement

Many small to medium libraries still rely on manual record keeping or basic spreadsheet systems, which lead to:

- Data inconsistency and redundancy
- Difficulty in tracking issued and returned books
- Time-consuming report generation
- Higher chances of human error
- Lack of centralized data management

BookHive addresses these problems by providing a secure, centralized, and user-friendly web-based solution for efficient library management.

3. Project Objectives

The objectives of the BookHive system are:

- To digitize library operations
- To simplify book and member management
- To ensure accurate tracking of issued and returned books
- To provide real-time statistics and reports
- To offer a scalable system for future enhancements

4. Scope of the System

- Librarian registration and login
- Book inventory management
- Member registration and management
- Issuing and returning books
- Membership application handling
- Report generation and data export
- Dashboard with statistics and charts

5. User Requirements Specification

5.1 Functional User Requirements

Librarian Authentication

- The system shall allow librarians to register using email and password
- The system shall authenticate librarians securely during login
- The system shall prevent duplicate email registrations

Book Management

- The librarian shall be able to add new books
- The librarian shall view all books in tabular form
- The librarian shall delete outdated or damaged books
- The system shall store book title, author, category, and quantity

Member Management

- The librarian shall register new library members
- The system shall store member personal and academic details
- The librarian shall update or delete member records

Book Issuing

- The librarian shall issue books to registered members
- The system shall record issue date and expected return date
- The system shall associate issued books with members and book records

Book Return

- The librarian shall record returned books
- The system shall store the condition of returned books
- The system shall maintain return history

Membership Application

- The system shall allow users to apply for library membership
- The system shall record membership type and duration

Reports and Analytics

- The system shall generate reports for books, members, issued and returned books
- The system shall allow data export in CSV, Excel, and PDF formats
- The system shall display charts and statistics on the dashboard

5.2 Non-Functional User Requirements

- The system shall be easy to use for non-technical users
- The system shall respond within acceptable time limits
- The system shall ensure data security and integrity
- The system shall be accessible via modern web browsers

6. System Requirements Specification (SRS)

6.1 Hardware Requirements

- Minimum 4 GB RAM
- Intel i3 processor or equivalent
- 10 GB free disk space

6.2 Software Requirements

- Operating System: Windows
- Web Server: Apache (XAMPP)
- Backend Language: PHP
- Database: MySQL
- Frontend: HTML5, CSS3, JavaScript, Bootstrap 5
- IDE: Visual Studio Code
- Browser: Google Chrome

7. Functional Requirements (System Perspective)

- Secure librarian login system
- CRUD operations for books and members
- Book issue and return tracking
- Membership application processing
- Report generation and export

- Dashboard with real-time statistics

8. Non-Functional Requirements (System Perspective)

8.1 Security

- Password hashing using password_hash()
- Input validation and sanitization
- Confirmation dialogs for destructive actions

8.2 Performance

- Optimized SQL queries
- Efficient data retrieval

8.3 Usability

- Responsive UI using Bootstrap
- Sidebar navigation and dashboard layout

8.4 Maintainability

- Modular PHP structure
- Centralized database connection file

9. System Architecture

BookHive follows a **three-tier architecture**:

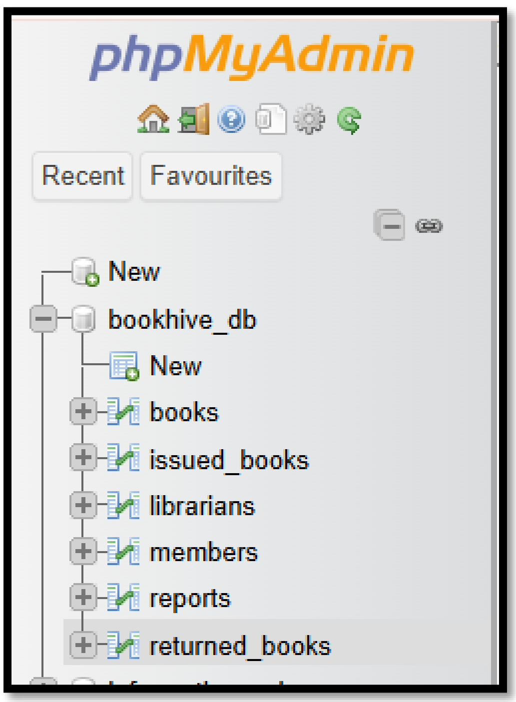
1. **Presentation Layer** – HTML, CSS, Bootstrap, JavaScript
2. **Application Layer** – PHP business logic
3. **Data Layer** – MySQL database

10. Database Design

Main Tables

- **librarians** (id, fullname, email, password)
- **books** (id, title, author, category, quantity)
- **members** (id, first name, last name,semester,course ,degree, email,phone,membership type, from date, to date,reason)
- **issued_books** (id, member_id, book_id, issue_date, return_date, remarks)
- **returned_books** (id, member_id, book_id, return_date, condition)

The database is normalized to reduce redundancy and ensure data consistency.



←T→	id	title	author	category	quantity
Edit Copy Delete	3	The Alchemist	Paulo Coelho	Fiction	2
Edit Copy Delete	4	Atomic Habits	James Clear	Self-improvement	3
Edit Copy Delete	5	Wren & Martin	P.C Wren & H.Martin	English Grammar	8
Edit Copy Delete	6	Oxford English Grammar Course	Michael Swan	English Language	9

	id	member_id	book_id	issue_date	return_date	remarks	created_at
Edit Copy Delete	3	4	3	2025-09-05	2025-09-19		2025-12-14 14:54:00
Edit Copy Delete	4	5	4	2025-09-10	2025-09-24		2025-12-14 14:54:35
Edit Copy Delete	5	6	5	2025-09-15	2025-09-29		2025-12-14 14:55:15

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Edit Copy Delete	2	Almas	23-20911-006@cs.fjwu.edu.pk	$2y$10$PhTi0WoJ4IC1HN./CZTUkutjzWlli9kb3pBo./UmV2e...
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4	hina	saad	1st	English	Bs English	23-20911-077@cs.fjwu.edu.pk	01324466768	Regular	2025-12-13	2025-12-30	volunteer work
5	Ayesha	Khan	3rd	Database Systems	BS Computer Science	ayesha.khan@example.com		Regular	2025-09-01	2026-09-01	Library access for semester study
6	Ali	Ahmed	5th	Operating Systems	BS Software Engineering	ali.ahmed@example.com		Volunteer	2025-08-15	2025-02-15	Research and project work

	id	member_id	book_id	return_date	condition	created_at
Edit Copy Delete	2	6	5	2025-09-29	Lost Pages	2025-12-14 14:57:15
Edit Copy Delete	3	5	4	2025-09-24	Good	2025-12-14 14:58:14

10.1 Code Libraries, Tools & Key Code Components Used

This section documents the libraries, functions, and code components used in the BookHive project, based on the implementation.

Frontend Libraries

- **Bootstrap 5**
  Used for responsive layout, forms, tables, buttons, modals, and dashboard UI.
  - Classes used: container, row, col-*, btn, table, modal, alert

- **JavaScript (Vanilla JS)**
  Used for:
  - Form validation
  - Page redirections and UI interactions
- **Chart.js**
  Used to display graphical statistics on the dashboard:
  - Total books
  - Issued books
  - Returned books

Backend Libraries / PHP Functions

- **mysqli (PHP extension)**
  Used for database connectivity and queries.
  - mysqli_connect()
  - mysqli_query()
  - mysqli_fetch_assoc()
  - mysqli_num_rows()
- **Password Security Functions**
  - password_hash() – for secure password storage
  - password_verify() – for login authentication
- **Form & Input Handling**
  - $_POST – form data handling
  - $_SESSION – session management after login
  - mysqli_real_escape_string() – input sanitization

Database & Server Tools

- **XAMPP** – Local server environment
- **Apache Server** – Web server
- **phpMyAdmin** – Database management

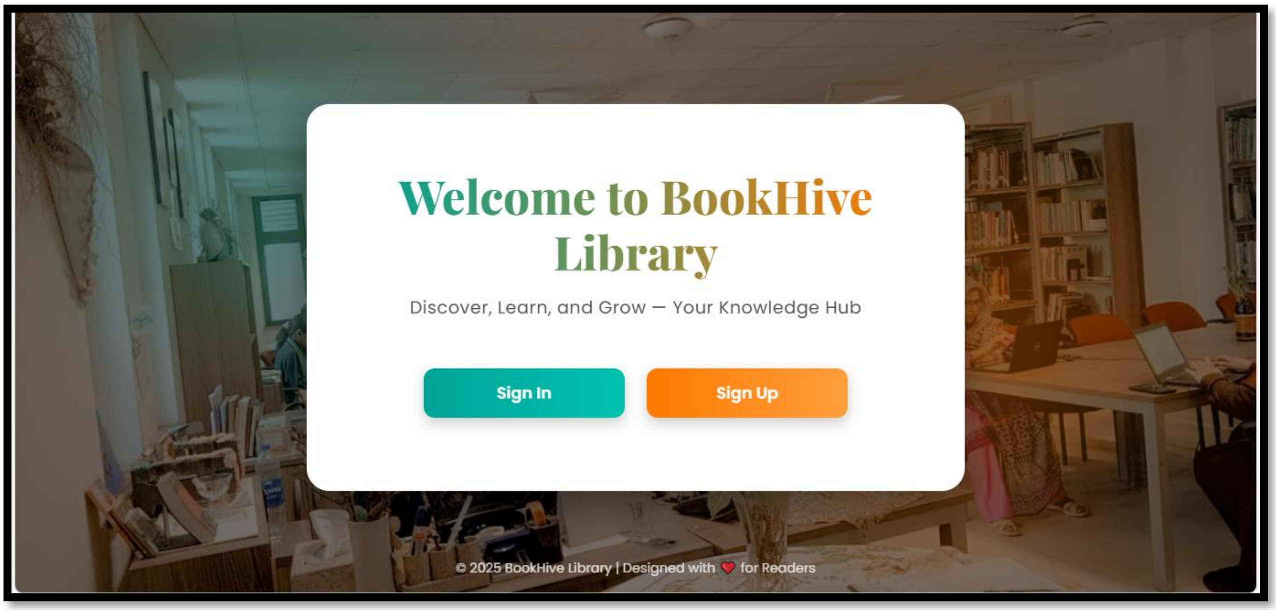
Important Code Files (Project Structure)

- db.php – Centralized database connection file
- signup.php – Librarian registration logic
- login.php – Authentication logic
- dashboard.php – Statistics and charts
- book.php – Book insertion and deletion
- members.php – Member management
- issue_book.php – Book issuing logic
- return_book.php – Book return logic
- repot.php – CSV / Excel / PDF report generation

11. User Interface Description

- Splash screen with animation
- Login and signup forms
- Sidebar-based dashboard
- Modal-based data entry forms
- Tabular data display with actions
- Charts using Chart.js


WELCOME TO BOOKHIVE


Welcome to BookHive Library
Discover, Learn, and Grow — Your Knowledge Hub
Sign In
Sign Up
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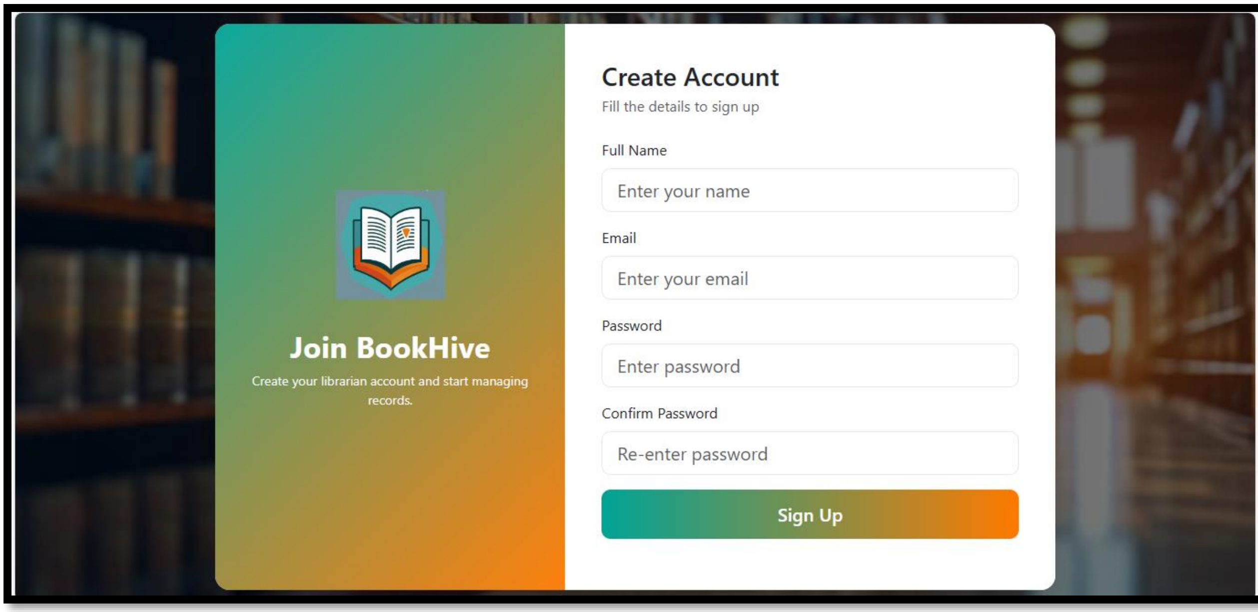
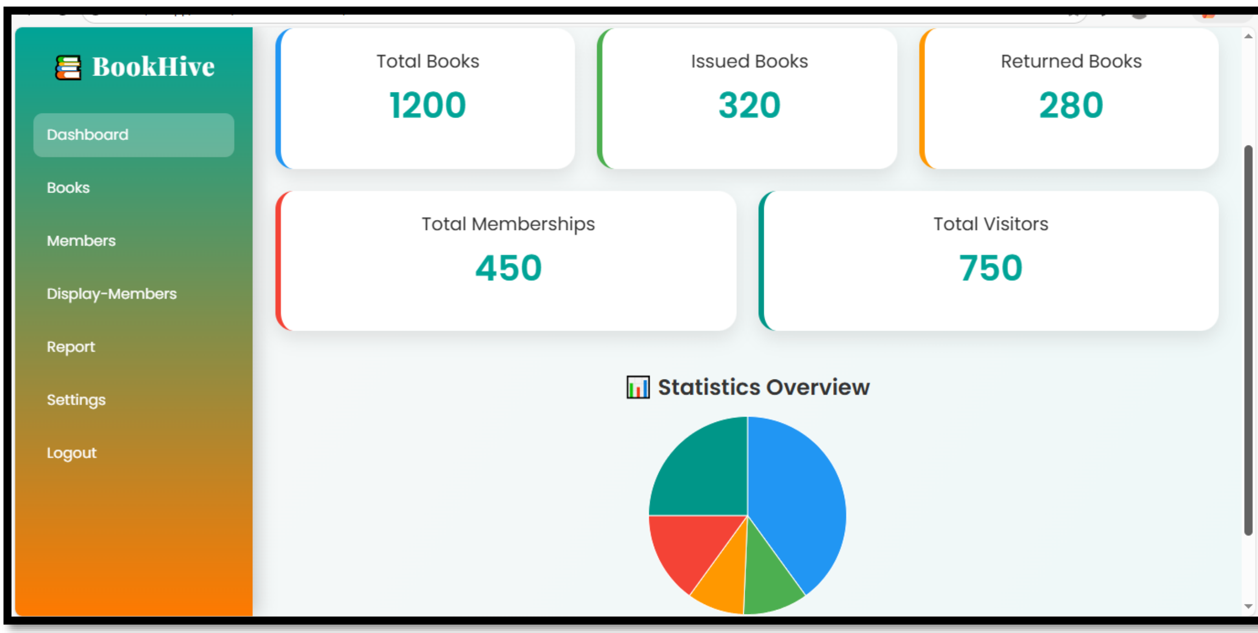

Join BookHive
Create your librarian account and start managing records.
Create Account
Fill the details to sign up
Full Name
Enter your name
Email
Enter your email
Password
Enter password
Confirm Password
Re-enter password
Sign Up


BookHive
Dashboard
Books
Members
Display-Members
Report
Settings
Logout
Total Books
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Issued Books
320
Returned Books
280
Total Memberships
450
Total Visitors
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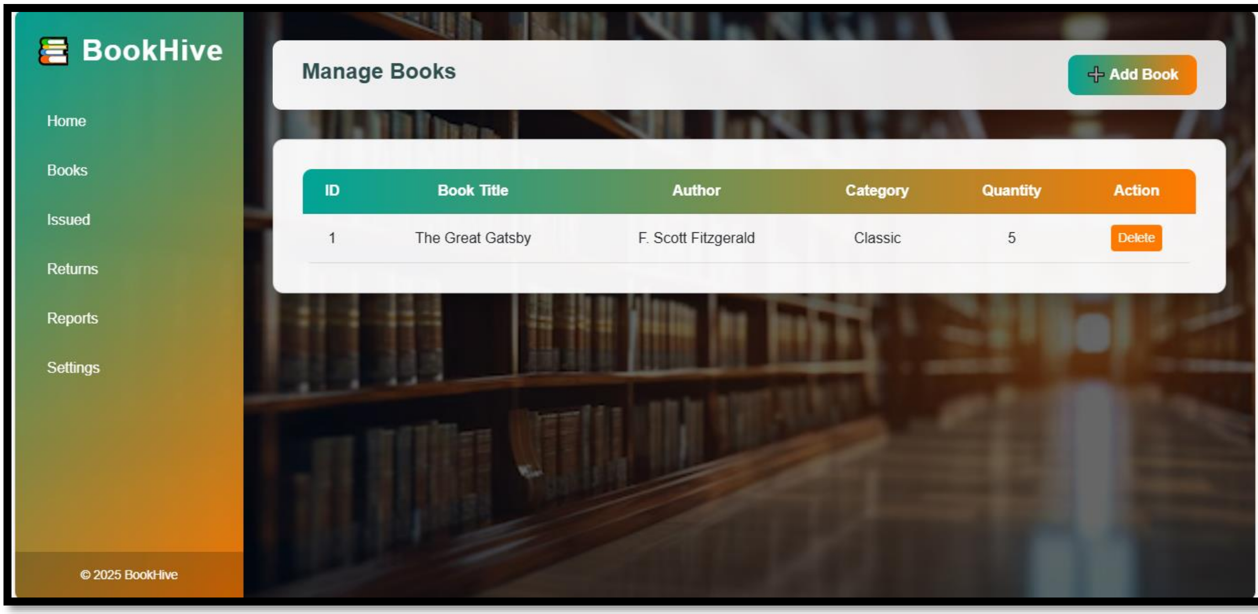
Statistics Overview



BookHive
Home
Books
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Settings
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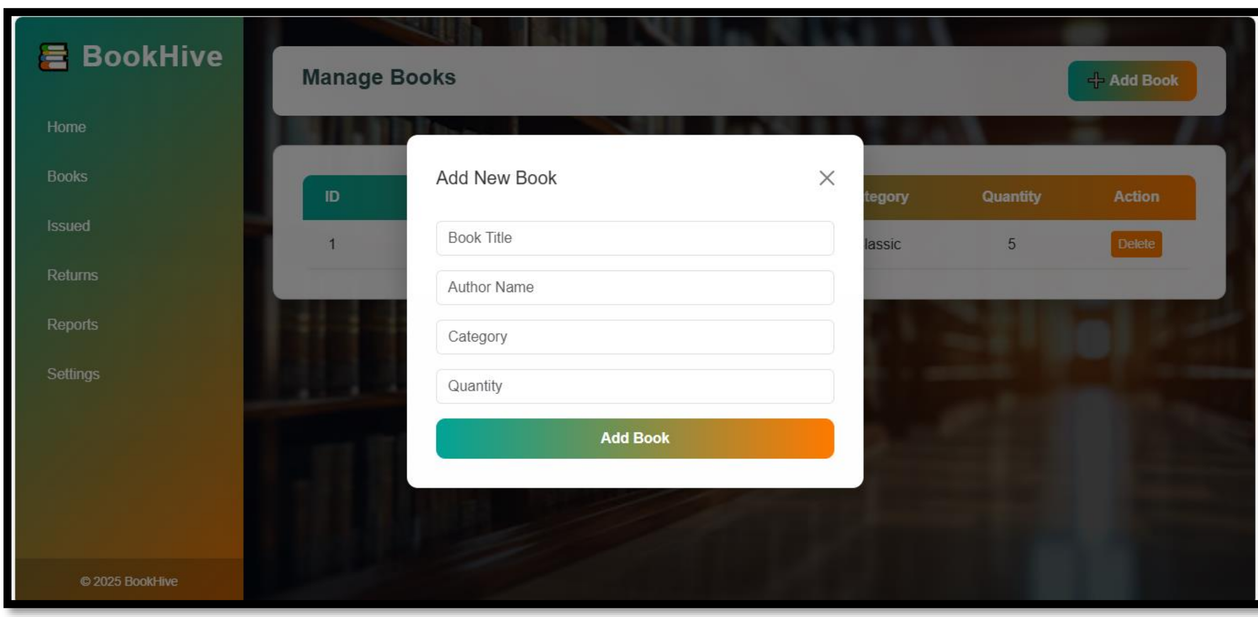

BookHive
Home
Books
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Returns
Reports
Settings
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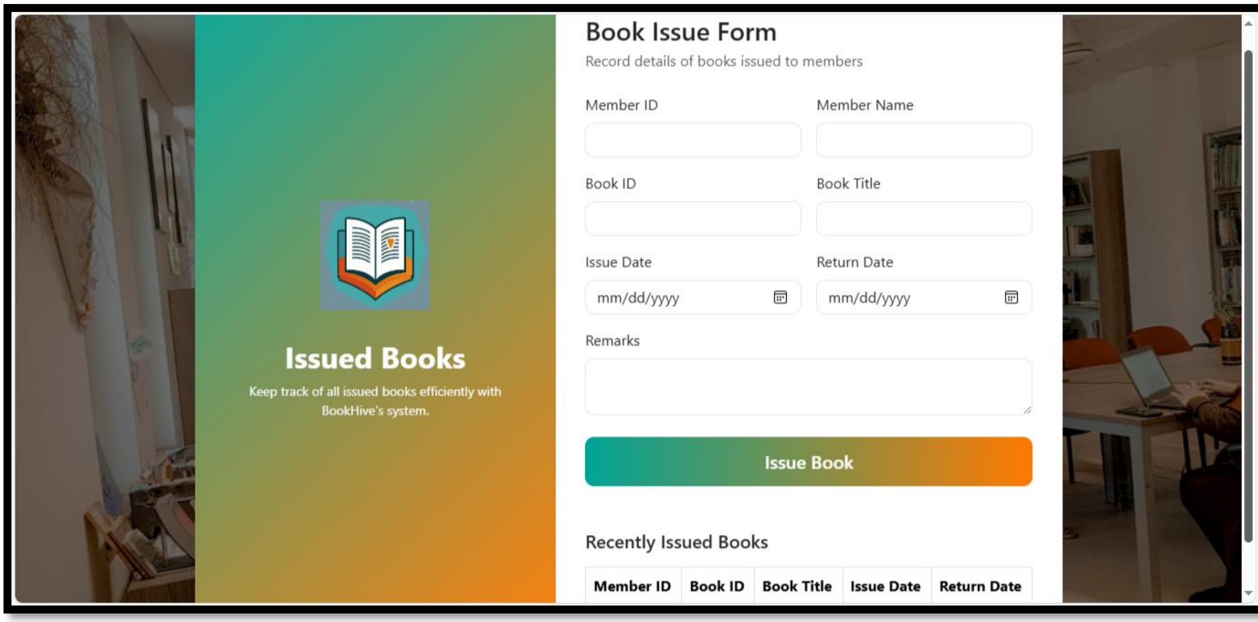
Manage Books
Add Book
Add New Book
Book Title
Author Name
Category
Quantity
Add Book


Issued Books
Keep track of all issued books efficiently with BookHive's system.
Book Issue Form
Record details of books issued to members
Member ID
Member Name
Book ID
Book Title
Issue Date
Return Date
mm/dd/yyyy
mm/dd/yyyy
Remarks
Issue Book
Recently Issued Books
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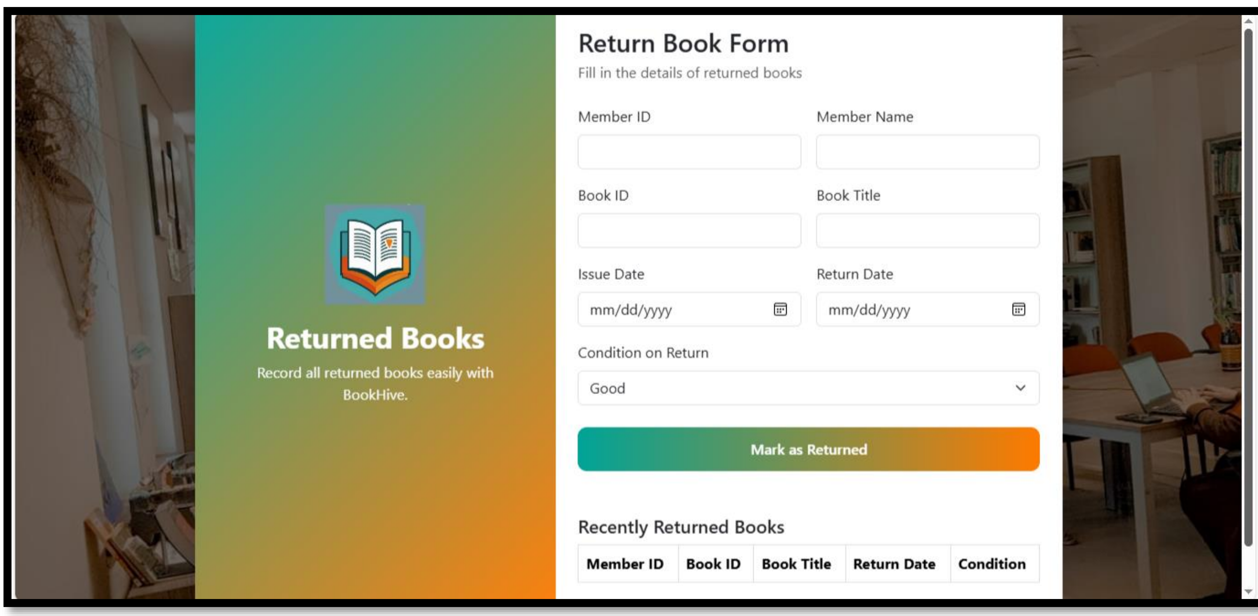
Book ID
Book Title
Issue Date
Return Date


Returned Books
Record all returned books easily with BookHive.
Return Book Form
Fill in the details of returned books
Member ID
Member Name
Book ID
Book Title
Issue Date
Return Date
mm/dd/yyyy
mm/dd/yyyy
Condition on Return
Good
Mark as Returned
Recently Returned Books
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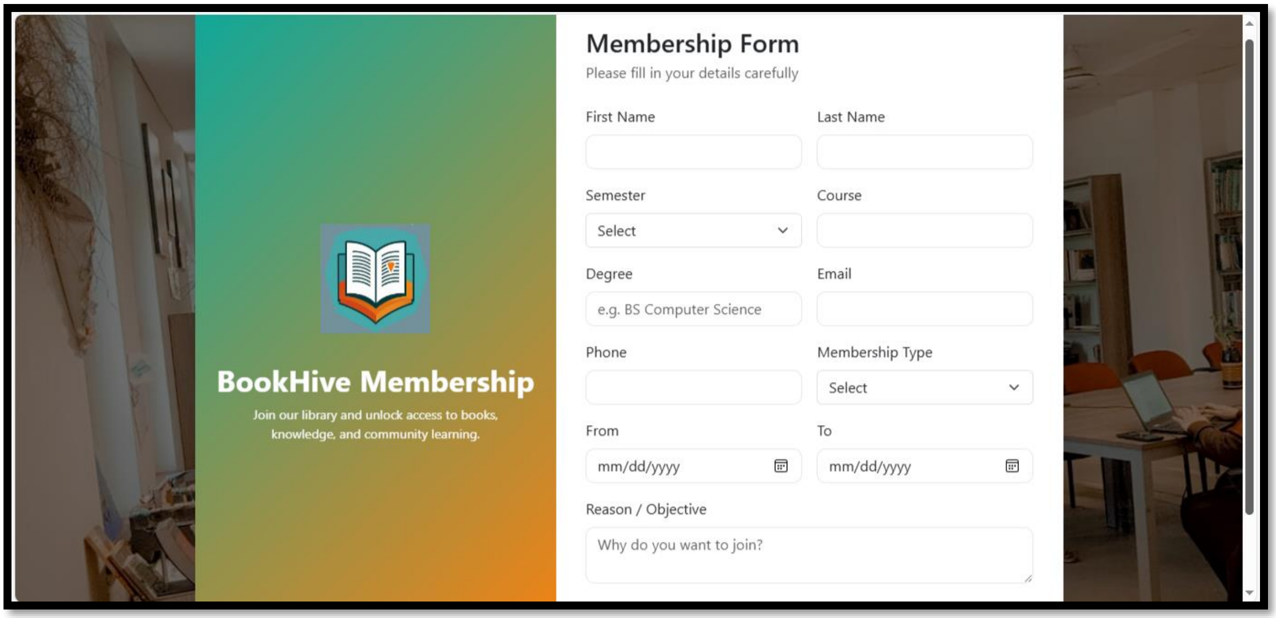
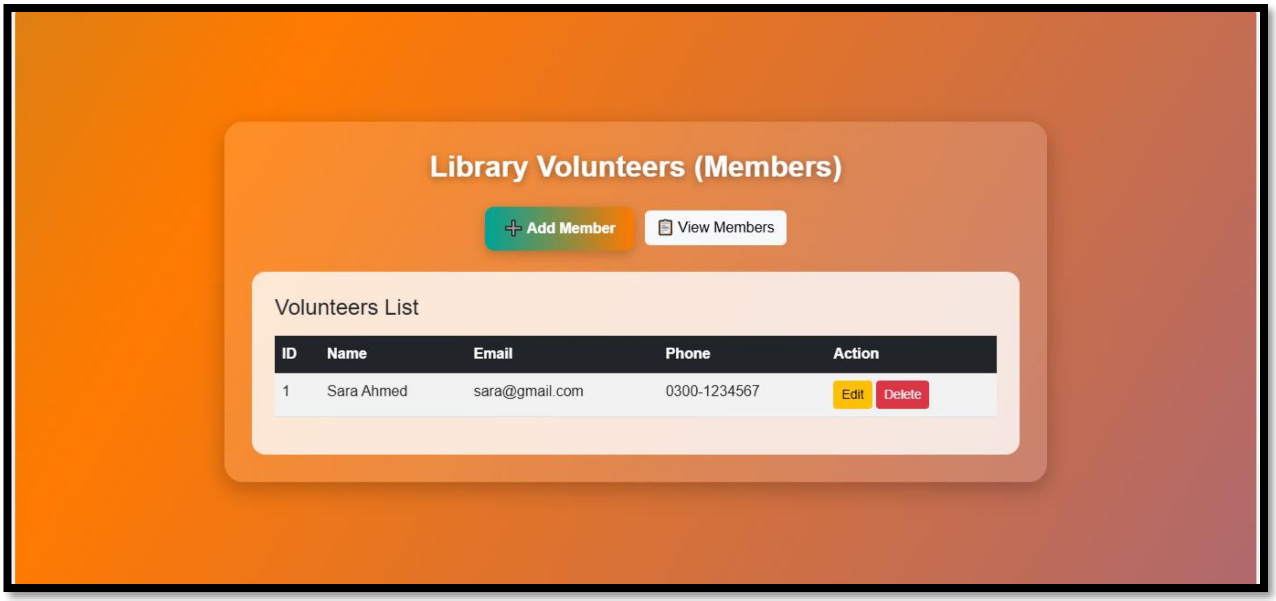
Return Date
Condition


BookHive Membership
Join our library and unlock access to books, knowledge, and community learning.
Membership Form
Please fill in your details carefully
First Name
Last Name
Semester
Select
Course
Degree
e.g. BS Computer Science
Email
Phone
Membership Type
Select
From
mm/dd/yyyy
To
mm/dd/yyyy
Reason / Objective
Why do you want to join?


Library Volunteers (Members)
Add Member
View Members
Volunteers List
ID
Name
Email
Phone
Action
1
Sara Ahmed
sara@gmail.com
0300-1234567
Edit
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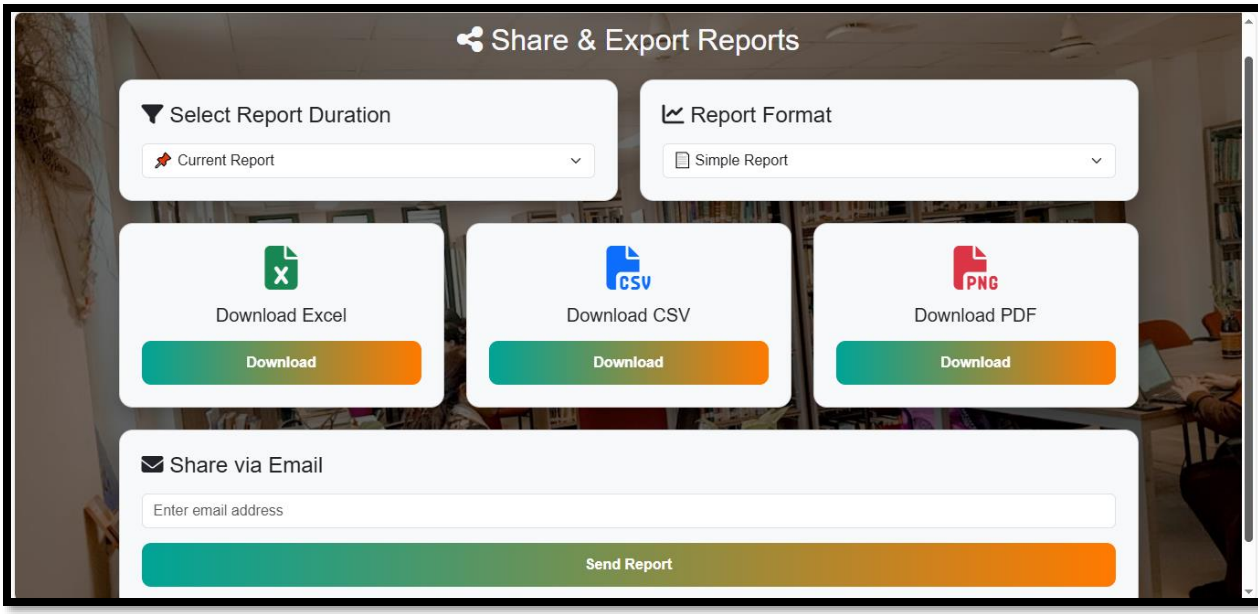

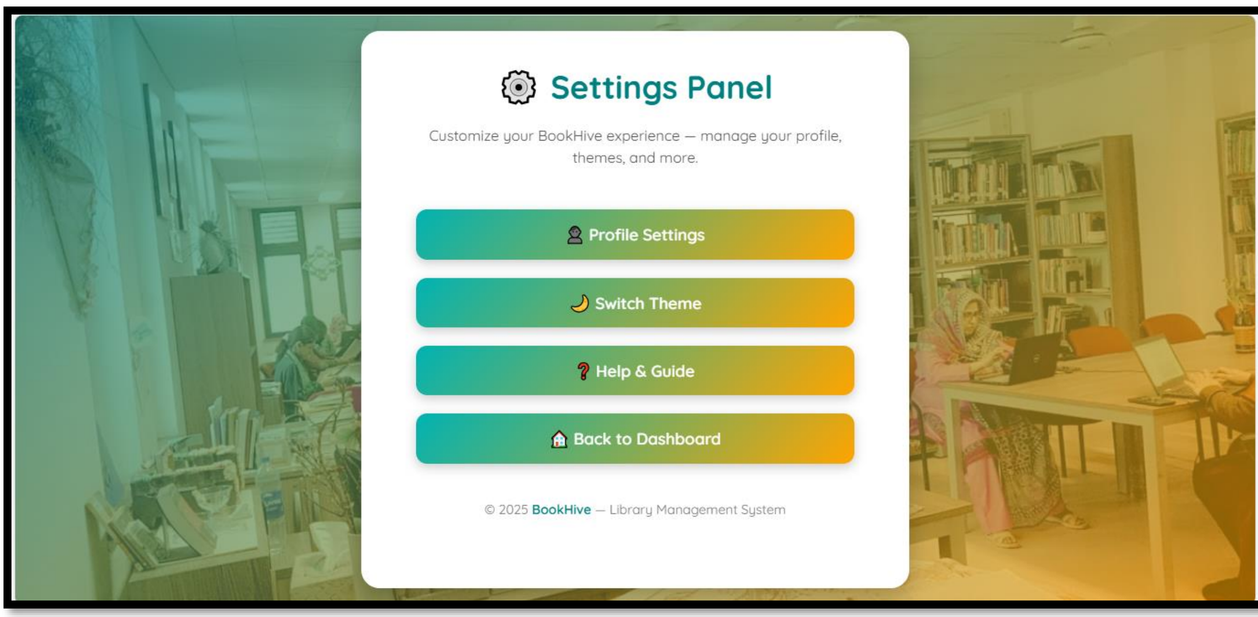

12. Validation and Error Handling

- Client-side validation using HTML5
- Server-side validation using PHP
- Error messages for invalid input
- Success alerts for completed actions

13. Testing Strategy

- Unit testing for individual modules
- Functional testing for CRUD operations
- Integration testing with database
- UI responsiveness testing

14. Technical Manual (Admin / Librarian)

14.1 System Setup

- Install XAMPP
- Place project folder in htdocs
- Import database using phpMyAdmin
- Configure database connection in db.php

14.2 Operating Instructions

- Login as librarian
- Use dashboard to navigate modules
- Add and manage books and members
- Issue and return books
- Generate reports

14.3 Maintenance Guidelines

- Regular database backup
- Secure admin credentials
- Update PHP and server regularly

15. Limitations

- Single role system (Librarian only)
- No fine management
- No email notifications
- Local deployment only

16. Future Enhancements

- Role-based access control
- Fine and payment management
- Cloud-based deployment

17. Conclusion

BookHive is a complete web engineering-based library management solution that fulfills user and technical requirements effectively. The project demonstrates strong understanding of requirement engineering, system design, implementation, and documentation, making it suitable for academic and real-world adaptation.